



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO VEYESS & CO.

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

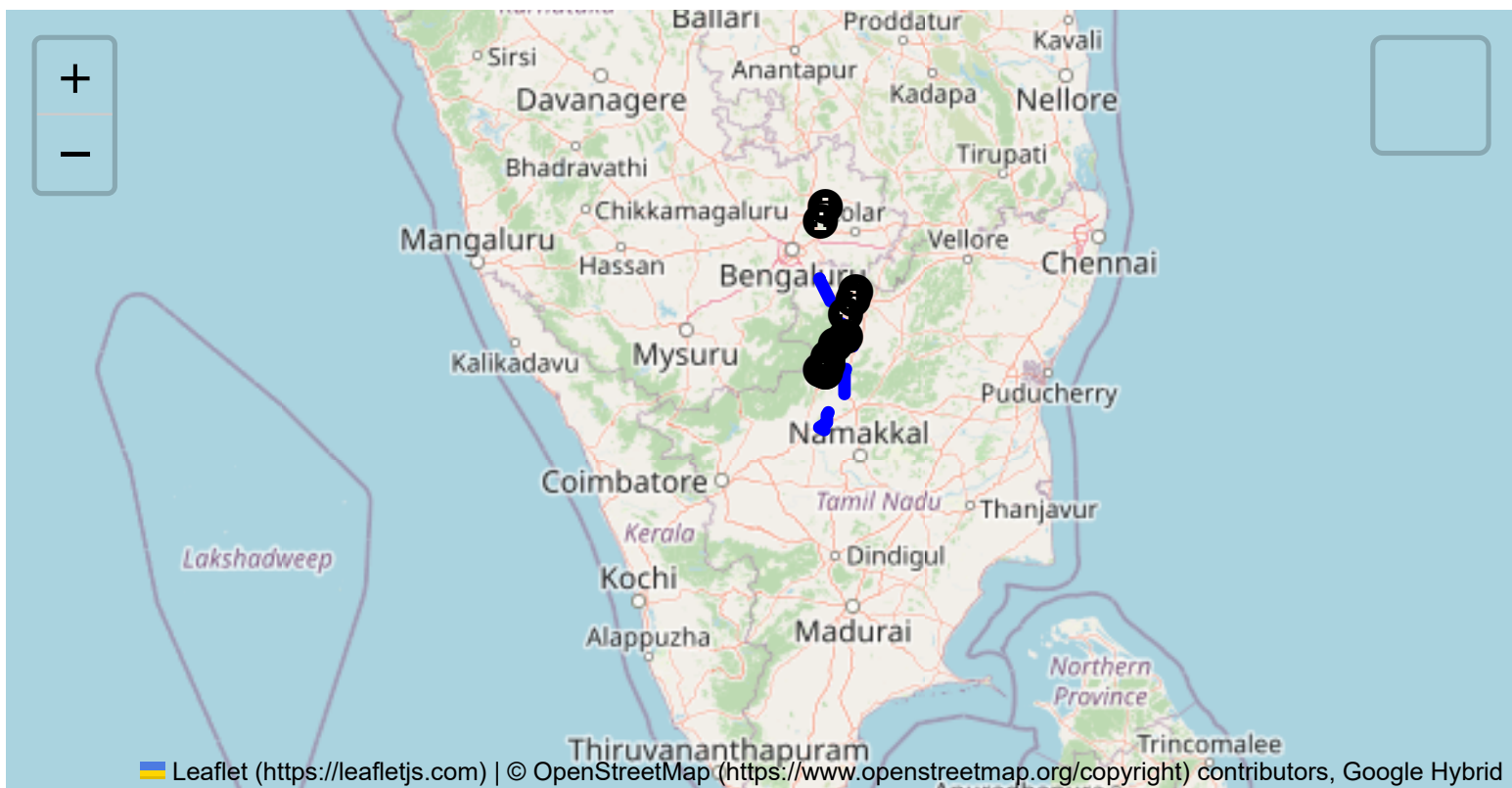
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

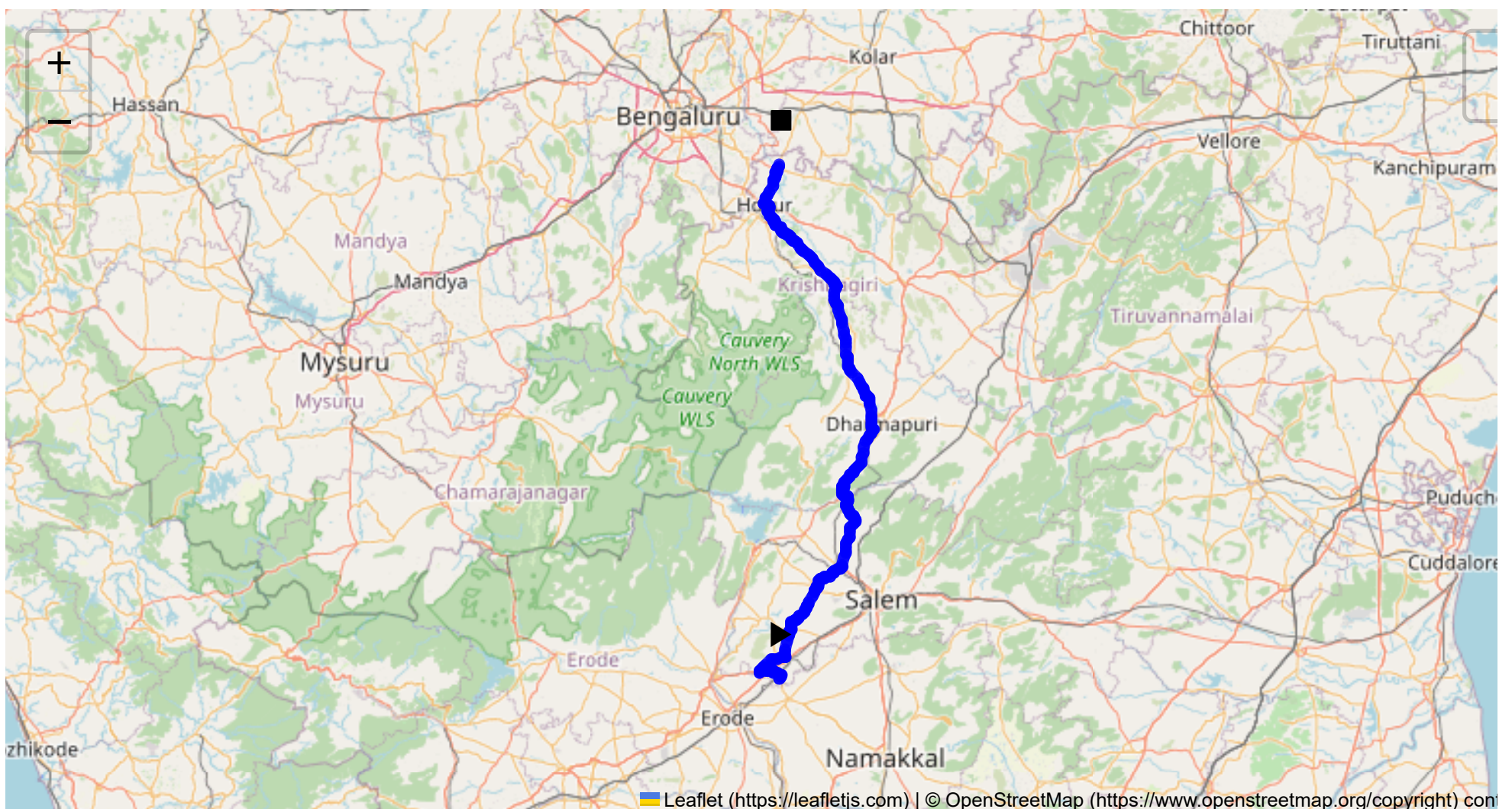
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 198.03 km
Estimated Duration: 4.2 hours
Adjusted Duration (Heavy Vehicle): 5.2 hours
Start: (11.4381, 77.8734)
End: (12.844593, 77.871727)



Welcome to the Journey Risk Management Study

1. Overview of the Route Map:

- The route spans approximately 198.03 kilometers, beginning at Sangagiri, Tamil Nadu, and ending at Bharathi Nagar, Kadiriganadinna, Tamil Nadu. It passes through multiple locations including Puthur, Padaiveedu, and Omalur. The primary highways used include the Salem - Kochi Highway and various state highways like SH 86.

2. Typical Weather Conditions and Potential Weather-Related Hazards:

- Tamil Nadu experiences a tropical climate, with the northeast monsoon (October to December) bringing heavy rains that could cause flooding and waterlogging. The region is also prone to heatwaves during summer (March to May), which might lead to road surface deterioration or heat-related vehicular issues.

3. Analysis of Traffic Patterns:

- The route typically faces heavy traffic congestion during morning (7 AM - 10 AM) and evening (5 PM - 8 PM) peak hours, especially in urban areas such as Salem and Hosur. National and state highways might also have heavy truck traffic consistently throughout the day.

4. Assessment of Road Quality and Infrastructure:

- Generally, the main highways like Salem - Kochi Highway are well-maintained, but some rural roads might have potholes or uneven surfaces. Construction activity and toll booths could also slow down traffic flow and increase travel time.

5. Suggestions for Alternative Routes for Emergencies:

- In case of roadblocks or severe congestion, alternative routes include using NH 44 or detours through nearby smaller towns which might not be optimal for heavy vehicles due to their narrow widths and lack of proper infrastructure.

6. Summary of Local Regulations Affecting Hazardous Material Transport:

- Tamil Nadu follows stringent regulations regarding the transport of hazardous materials, requiring proper labeling, safety documentation, and adherence to specific routes designated for hazardous material transport to minimize risk to populated areas.

7. Overview of Historical Incidents Involving Heavy Vehicles:

- Past incidents have typically involved vehicular collisions or overturning due to speeding on curves, especially in hilly or poorly lit sections of the route. Specific data may be drawn from local transport and safety reports.

8. Environmental Considerations and Sensitive Areas:

- The route passes near ecologically sensitive areas like reserved forests which require extra caution to avoid polluting these regions. Local wildlife crossings are also important to monitor to prevent accidents.

9. Analysis of Communication Coverage:

- While most of the route is covered by major cellular networks, rural and forested stretches might feature dead zones or weak signals impacting communication, especially in areas like Desavilakku and parts near Thoppur.

10. Estimated Emergency Response Times:

- In urban areas, emergency response times are approximately 15-30 minutes, whereas in rural or remote areas, response times may extend to an hour or more due to limited resources and accessibility challenges.

11. Overall Summary of Risk Assessment:

- The route from Sangagiri to Bharathi Nagar poses moderate risks for heavy vehicles carrying hazardous materials. Key concerns include weather-related conditions, road quality variances, and traffic congestion. Adequate route planning, adherence to local transportation laws, and real-time monitoring for emergency preparedness can mitigate risks effectively. Regular checks on vehicle condition and driver alertness are imperative, particularly through environmentally sensitive and remote stretches.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	12.83040, 77.86626	15 KM/Hr
1	Turn	High	11.43961, 77.87341	15 KM/Hr
2	Turn	High	11.44077, 77.87533	15 KM/Hr
3	Turn	High	11.44898, 77.87410	15 KM/Hr
4	Turn	Medium	11.45352, 77.85698	30 KM/Hr
5	Turn	Medium	11.45764, 77.85219	30 KM/Hr
6	Turn	High	11.46303, 77.84945	15 KM/Hr
7	Turn	Medium	11.45545, 77.81812	30 KM/Hr
8	Turn	High	11.45584, 77.81695	15 KM/Hr
9	Turn	Medium	11.45656, 77.81709	30 KM/Hr
10	Turn	High	11.49383, 77.88442	15 KM/Hr
11	Turn	Medium	11.56352, 77.90078	30 KM/Hr
12	Turn	Medium	11.59189, 77.90645	30 KM/Hr
13	Turn	Medium	11.59328, 77.90809	30 KM/Hr
14	Turn	Medium	11.63867, 77.94841	30 KM/Hr
15	Turn	High	11.64217, 77.95200	15 KM/Hr
16	Turn	Medium	11.64712, 77.95118	30 KM/Hr
17	Turn	Medium	11.67858, 77.96834	30 KM/Hr
18	Turn	Medium	11.69776, 77.97983	30 KM/Hr
19	Turn	Medium	11.70179, 77.97900	30 KM/Hr
20	Turn	Medium	11.71394, 77.99572	30 KM/Hr
21	Turn	Medium	11.71916, 78.01117	30 KM/Hr
22	Turn	Medium	11.72504, 78.01680	30 KM/Hr
23	Turn	Medium	11.73766, 78.03445	30 KM/Hr
24	Turn	High	11.73651, 78.03868	15 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
25	Turn	Medium	11.74120, 78.04588	30 KM/Hr
26	Turn	Medium	11.86805, 78.08353	30 KM/Hr
27	Turn	Medium	11.87913, 78.08206	30 KM/Hr
28	Turn	Medium	11.91159, 78.05857	30 KM/Hr
29	Turn	Medium	11.92566, 78.06503	30 KM/Hr
30	Turn	Medium	11.93307, 78.06426	30 KM/Hr
31	Turn	Medium	11.93545, 78.06130	30 KM/Hr
32	Turn	Medium	11.93694, 78.05425	30 KM/Hr
33	Turn	Medium	11.95531, 78.04698	30 KM/Hr
34	Turn	High	11.95803, 78.04964	15 KM/Hr
35	Turn	Medium	11.96068, 78.04960	30 KM/Hr
36	Turn	High	11.96370, 78.05142	15 KM/Hr
37	Turn	High	11.96650, 78.04927	15 KM/Hr
38	Turn	Medium	12.12094, 78.13269	30 KM/Hr
39	Turn	High	12.51391, 78.03197	15 KM/Hr
40	Turn	High	12.51369, 78.02985	15 KM/Hr
41	Turn	Medium	12.51594, 78.02617	30 KM/Hr
42	Turn	Medium	12.52198, 78.02399	30 KM/Hr
43	Turn	Medium	12.52311, 78.02280	30 KM/Hr
44	Turn	Medium	12.52966, 78.01952	30 KM/Hr
45	Turn	Medium	12.60846, 77.93721	30 KM/Hr
46	Turn	Medium	12.61024, 77.93626	30 KM/Hr
47	Turn	Medium	12.61129, 77.93385	30 KM/Hr
48	Turn	Medium	12.65815, 77.88249	30 KM/Hr
49	Turn	Medium	12.67924, 77.86193	30 KM/Hr
50	Turn	Medium	12.68946, 77.85539	30 KM/Hr
51	Turn	Medium	12.69429, 77.85425	30 KM/Hr
52	Turn	Medium	12.71847, 77.83804	30 KM/Hr
53	Turn	High	12.71955, 77.83557	15 KM/Hr
54	Turn	High	12.71966, 77.83553	15 KM/Hr
55	Turn	Medium	12.72295, 77.83931	30 KM/Hr
56	Turn	Medium	12.72545, 77.84038	30 KM/Hr
57	Turn	Medium	12.72645, 77.84216	30 KM/Hr
58	Blind Spot	Blind Spot	12.72811, 77.84326	10 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
59	Turn	High	12.73513, 77.82950	15 KM/Hr
60	Turn	Medium	12.73995, 77.83287	30 KM/Hr
61	Turn	Medium	12.80987, 77.85779	30 KM/Hr
62	Turn	Medium	12.82863, 77.86586	30 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
1	hospital	Sri Shanthi Hospital	11.569272, 77.901937	30 km/h	Medium
2	hospital	Omalur Government Hospital	11.74161, 78.044296	30 km/h	Medium
3	clinic	R.R. Nursing Home	11.7394129, 78.0439861	30 km/h	Medium
4	hospital	Sri Hari Hospital	11.7396726, 78.0440701	30 km/h	Medium
5	hospital	Sugumar Hospital	11.7396031, 78.0426955	30 km/h	Medium
6	hospital	Sri Sugam Hospital	11.7429821, 78.0461039	30 km/h	Medium
7	hospital	Saravana Hospital	11.7409303, 78.0458308	30 km/h	Medium
8	hospital	Arun Hospital	11.7429892, 78.0466881	30 km/h	Medium
9	clinic	Saratha Clinic	11.7422818, 78.0458175	30 km/h	Medium
11	clinic	Sri Meenakshi Nursing Home, Salem	11.7449162, 78.0474637	30 km/h	Medium
12	hospital	Vinayaka Child Hospital	11.745812, 78.0480225	30 km/h	Medium
13	hospital	Dr. G.R.Thilakam Hospital	11.746097, 78.04769	30 km/h	Medium
14	clinic	Divyam Child Care Clinic	11.7447135, 78.0473858	30 km/h	Medium
15	hospital	Thulasi Hospital	11.7461516, 78.0469984	30 km/h	Medium
16	clinic	Momota Clinic	11.744982, 78.047397	30 km/h	Medium
17	hospital	Amirthalingam Hospital	11.7456708, 78.0470548	30 km/h	Medium
18	hospital	Children Hospital	11.7458377, 78.0480205	30 km/h	Medium
19	clinic	Gomathi Clinic	11.7445293, 78.0470401	30 km/h	Medium
20	clinic	Manimuthu Skin Clinic	11.7521189, 78.0484058	30 km/h	Medium
22	hospital	L.R. Hospital	11.8703354, 78.0844554	30 km/h	Medium
23	clinic	JSA Clinic	11.8700619, 78.0844982	30 km/h	Medium
29	pharmacy	H2 Pharmacy	12.7301802, 77.8401037	30 km/h	Medium
30	clinic	Meera Nursing Home, Krishnagiri	12.7326185, 77.8292795	30 km/h	Medium
31	hospital	Amma Hospital	12.7393554, 77.8326981	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
34	pharmacy	துளசி பார்மசி	12.7457796, 77.8339248	30 km/h	Medium
36	clinic	Thirumala Private Nursing Home	12.835671, 77.8672146	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	school	KRP Matric. Hr. Sec School	11.4546193, 77.8142445	30 km/h	Medium
10	school	Fathima Hr Sec School	11.7410508, 78.0479272	30 km/h	Medium
21	college	Dhirajlal Gandhi College of Technology	11.7968014, 78.0562095	30 km/h	Medium
24	college	Laxminarayana Polytechnic College	12.0421277, 78.103203	30 km/h	Medium
25	marketplace	Rayakottai tomato market	12.5237469, 78.0220377	30 km/h	Medium
26	school	Government Higher Secondary School	12.5844162, 77.9669764	30 km/h	Medium
27	school	Panchayat Union Primary School	12.6078768, 77.9381865	30 km/h	Medium
28	school	Panchayat Union Primary School	12.6835154, 77.8604798	30 km/h	Medium
32	school	Vijay Vidyalaya School	12.7397242, 77.8308427	30 km/h	Medium
33	school	Maharishi Vidhya Mandir Primary School	12.7397254, 77.8324166	30 km/h	Medium
35	school	யோகி வேமனா உயர்நிலை பள்ளி	12.7952478, 77.8563035	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 12.83040, 77.86626



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44077, 77.87533



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44898, 77.87410



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45352, 77.85698



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45764, 77.85219



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.46303, 77.84945



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45545, 77.81812



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45584, 77.81695



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45656, 77.81709



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49383, 77.88442



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.56352, 77.90078



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.59189, 77.90645



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.59328, 77.90809



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.63867, 77.94841



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.64217, 77.95200



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.64712, 77.95118



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.67858, 77.96834



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.69776, 77.97983



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.70179, 77.97900



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.71394, 77.99572



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.71916, 78.01117



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.72504, 78.01680



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.73766, 78.03445



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.73651, 78.03868



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.74120, 78.04588



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.86805, 78.08353



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.87913, 78.08206



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.91159, 78.05857



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.92566, 78.06503



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.93307, 78.06426



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.93545, 78.06130



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.93694, 78.05425



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.95531, 78.04698



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.95803, 78.04964



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96068, 78.04960



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.96370, 78.05142



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.96650, 78.04927



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.51391, 78.03197



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.51369, 78.02985



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.51594, 78.02617



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.52198, 78.02399



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.52311, 78.02280



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.52966, 78.01952



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.60846, 77.93721



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.61024, 77.93626



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.61129, 77.93385



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.65815, 77.88249



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.67924, 77.86193



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.68946, 77.85539



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.69429, 77.85425



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.71847, 77.83804



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 12.71955, 77.83557



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.71966, 77.83553



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.72295, 77.83931



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.72545, 77.84038



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.72645, 77.84216



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 12.72811, 77.84326



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.73513, 77.82950



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.73995, 77.83287



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.80987, 77.85779



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.82863, 77.86586
